N-BPMSNet: An NDMI-Guided Bitemporal Network for Methane Plume Detection and Segmentation From Sentinel-2 Multispectral Observations

Di Xu, Philippa J. Mason, Jianguo Liu, *Senior Member, IEEE*, and Yanghua Wang

***Abstract*—Mitigating methane emissions is critical for addressing global warming, and accurate point-source detection is crucial for identifying super-emitters and targeted mitigation. With its fine spatial resolution, global coverage, and open accessibility, Sentinel-2 offers strong potential for large-scale methane monitoring. Current detection methods, whether traditional retrieval-based approaches or AI-based models, suffer from frequent false positives, low signal-to-noise ratios, reliance on manual verification, and limited generalization across heterogeneous surfaces, leading to unreliable plume identification and segmentation. To address these challenges, we propose normalized difference methane index-guided bi-temporal plume methane segmentation network (N-BPMSNet), a Sentinel-2-based methane plume detection and segmentation framework. The model integrates temporal, spectral, spatial, and differential information through specialized modules that guide the network to focus on subtle plume regions and accurately extract them from complex and interfering backgrounds. A large-scale real-world Sentinel-2 dataset from 44 point sources across the USA, Algeria, and Turkmenistan is also constructed, comprising 11 494 samples with annotated methane plumes and nonplume scenes across various land cover types and emission facilities for model training, validation, and evaluation. N-BPMSNet has been tested and achieved an $F1$-score of 0.8858 and an AUC of 0.9856 in the test dataset, outperforming existing approaches and advanced segmentation networks. Out-of-domain generalization is further evaluated using supplementary datasets from Khuzestan Province, Iran, and an O&G region in Algeria, demonstrating robust performance in unseen regions. Case studies confirm its robustness to confounding factors such as flare smoke, cloud shadows, and offshore emissions, while an inference speed on the order of 0.1 s per 3 × 3 km tile highlights its scalability for operational monitoring. Collectively, the experimental results demonstrate that N-BPMSNet achieves accurate, automated, and scalable plume segmentation under diverse environmental and surface conditions, making this method an important contribution to the detection of methane plumes from freely available data sources.**

***Index Terms*—Methane plume detection, multispectral imagery, segmentation, self-attention.**

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I. Introduction

METHANE, as a major anthropogenic greenhouse gas, has contributed approximately 20% of global warming [1]. With continuously increasing methane emissions, its impact on global warming may be as much as 84 times that induced by carbon dioxide over the last 20 years [2]. Therefore, effective reduction of methane emissions is a serious and urgent issue. Methane emissions arise from a variety of sources, such as wetlands, landfills, livestock and oil/gas production and consumption [3], but often exhibit a so-called "long-tail distribution," namely a small number of super-emitters contribute significantly to overall emissions according to the research so far [4], [5], [6], [7], [8], [9]. Therefore, accurate detection and quantification of global anomalous point sources of methane emissions, which typically produce plumes with a horizontal extent below 1 km [4], [5], [10], are crucial to effectively monitoring and controlling, ultimately helping to mitigate greenhouse warming.

In the latest studies, remotely sensed datasets from Earth Observation satellites and aircraft, have demonstrated powerful capabilities in the detection and quantification of atmospheric methane, based on methane's spectral properties, especially its strong absorption bands at 1650 and 2300 nm in the shortwave infrared (SWIR) spectrum [11], [12], [13], [14]. The Sentinel-5P TROPOMI satellite (launched in 2017) [15] provides global continuous daily coverage of hyperspectral imagery with 0.25 nm spectral resolution; however, it lacks the capacity to locate point sources and detect small emissions due to coarse spatial resolutions ranging from 3.5×7 km^2 to 7×7 km^2, depending on the spectral band [16], [17]. Hyperspectral satellite instruments, such as EnMAP [18], PRISMA [19], EMIT [20], and GHGSat [21], offer finer spatial resolution images around 30 m and lower detection thresholds, with the estimated capability of detecting small methane point sources with emissions $\sim$ 0.1 t_{CH4}/h with relatively high precision of 3%–15% [21], [22], [23]. However, due to limited spatial coverage, these satellites are unable to monitor the global distribution of methane point sources [10], [24]. Airborne instruments such as AVIRIS-NG [25] have even better spatial

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resolutions of ~5 m and lower detection thresholds (below 0.01 t_{CH4}/h) but they cannot easily provide routine, global data collection [26], [27], [28]. Although they can achieve the detection of smaller point sources, airborne campaigns are only suitable for well-targeted areas with very constrained temporal and spatial coverage. In comparison, multispectral satellite platforms, such as Sentinel-2, provide significant advantages in tracking large methane point sources overextended periods and in global scope because of their finer spatial resolution (10–60 m global coverage, and short revisit periods of 2–5 days [29]). Notably, its two SWIR bands, Band 11 and Band 12, are sensitive to the spectral characteristics of methane absorption, enabling the identification and quantification of point-source methane emissions [10], [17], [24], [30].

Despite these advantages, existing approaches for accurate detection and segmentation of methane plumes from Sentinel-2 imagery have several significant limitations. The first and most critical issue is the frequent false positives and low signal-to-noise ratio (SNR), primarily due to Sentinel-2's limited spectral resolution. Many surface features, such as clay-rich soils, vegetation, water bodies, and anthropogenic edge features, exhibit spectral similarities to methane in the SWIR, generating interference and leading to false detections. This is particularly true on heterogeneous land surfaces, where complex surface reflectance patterns exacerbate the problem. Several recent works have applied denoising techniques, including median filter and Gaussian filter based on Band Ratio [24], multiband-multipass (MBMP) retrieval [10], and Matched Filter [31] detection methods to eliminate artifacts. Additional strategies include the removal of outliers and normalization of the retrievals [32], as well as masking water bodies using indices such as MNDWI [33]. Although these methods can mitigate some of the interference and artifacts over relatively homogeneous surfaces, they may also introduce additional small-scale noise and false detections over complex surfaces. In addition, methods that directly mask water bodies or vegetation can compromise the detection of methane plumes over rural landscapes or offshore regions.

A second limitation is the requirement for manual verification and threshold adjustments for plume selection, which makes traditional approaches computationally intensive and inefficient for large-scale operational deployment [13], [34]. To address this, AI-based models have recently been introduced to attempt automated identification of methane plumes in remote sensing imagery. Several studies have explored plume classification and emission quantification using models such as MethaNet (Jongaramrungruang et al. [35]), Resnet (Zortea et al. [33]), S2MetNet (Radman et al. [36]) and DSAN (Zhao et al. [37]). However, relatively few of these focus on plume segmentation, which is critical to accurately separate spatial plume patterns from background pixels and further quantify emissions. This segmentation task remains particularly challenging due to the small spatial proportion and complex morphology of methane plumes in the vast background of the image. Within the domain of remote sensing, various deep learning segmentation architectures, such as UNet [38], SiamCRNN [39], SwinSUNet [40], and MambaBCD [41], have demonstrated strong capabilities in extracting spatial structures and contextual features. Building on these models, several approaches have been proposed for plume segmentation using hyperspectral imagery from PRISMA, GHGSat, and EMIT [42], [43], [44], while multispectral observations are limited by their coarser spectral resolution. Rouet-Leduc and Hulbert [45] applied a ViT model trained on synthetic plumes to address the lack of labeled real-world data. However, such simulations inherently constrain model generalization under complex real-world conditions, including varying atmosphere conditions, cloud cover, and surface reflectivity. Models trained on real-world data, such as CH4Net [46] and MultiSTARCOP [47], also face difficulties in distinguishing plumes from interfering surface features, largely due to limited coverage of heterogeneous land surfaces in training datasets and the reliance on single-date imagery. Consequently, these factors often result in frequent false positives and reduced generalization under diverse real-world conditions.

To overcome these limitations, we propose normalized difference methane index-guided bi-temporal plume methane segmentation network (N-BPMSNet), a novel normalized difference methane index (NDMI)-guided Bi-temporal Plume Methane Segmentation Network for accurate and automated methane plume detection and segmentation. The proposed network aims to reduce surface false positives, accelerate processing, and improve generalization in various surface scenarios. N-BPMSNet captures methane plumes from Sentinel-2 reflectance imagery.

Unlike existing CNN-based methane detection methods that primarily focus on feature extraction alone, the strength of N-BPMSNet lies in its joint exploitation of physically interpretable spectral indices and data-driven temporal learning. Specifically, the NDMI-guided architecture explicitly integrates methane-sensitive spectral features with bi-temporal inputs, which reflect the temporal variation of plumes, into the network learning process, bridging the gap between physical methane absorption principles and deep representation learning. This design not only improves the discrimination of weak plumes but also enhances generalization across heterogeneous surfaces and atmospheric conditions.

The key contributions of this work are summarized as follows.

1) *New Network Design and Modules:* The proposed N-BPMSNet incorporates spectral and temporal differences to capture dynamic plume behavior, a guided attention mechanism based on the NDMI image to enhance spectral sensitivity to methane, and an effective training strategy to balance the relatively small proportion of plumes to background pixels.
2) *Real-World Dataset Construction:* A comprehensive labeled real-world Sentinel-2 dataset with plume and nonplume annotations across various surface and atmospheric conditions and multiple years, supporting robust training, evaluation and generalization.
3) *Comprehensive Evaluation:* Extensive comparative experiments, ablation studies, visualizations, and case studies are conducted using a real-world Sentinel-2 dataset of 44 methane point sources in the USA,

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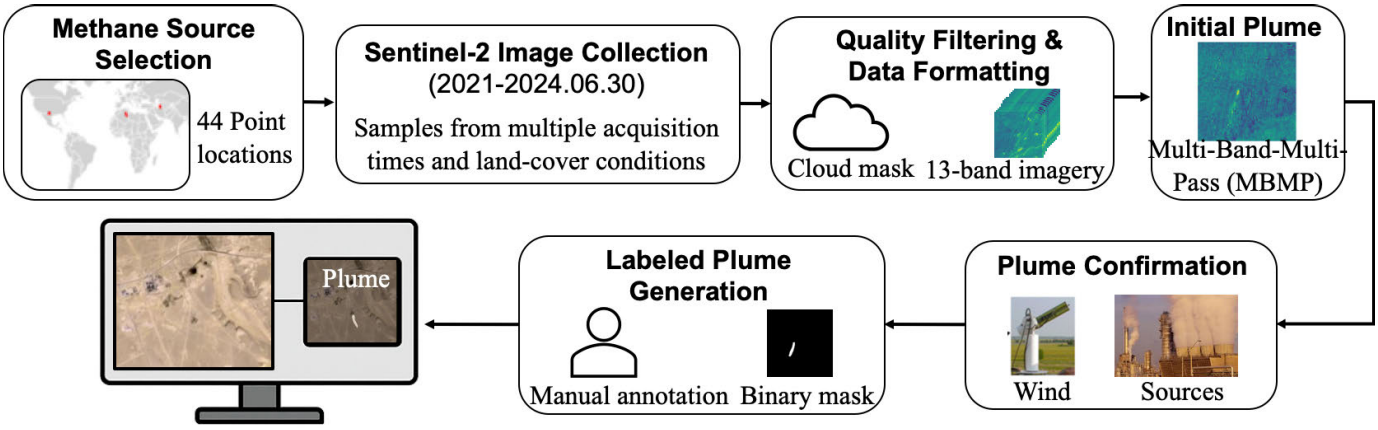


Fig. 1. Overview of the data collection and preprocessing pipeline for methane plume detection using sentinel-2 imagery. The process includes site selection, sentinel-2 L1C imagery collection, quality control, preliminary plume identification, cross-validation for plume confirmation, and manual annotation to generate binary plume masks.

Algeria, and Turkmenistan. To assess out-of-domain generalization, independent datasets from Khuzestan Province, Iran and an O&G region in Algeria (excluded from training, validation and testing) are also evaluated. Results demonstrate the effectiveness of N-BPMSNet against state-of-the-art methods, and its robustness to confounding factors such as flare smoke, cloud shadows, and offshore emissions, highlighting its potential for operational deployment in large-scale methane monitoring.

II. Data

A. Sentinel-2 Data

The Sentinel-2 mission, developed by the European Space Agency (ESA), provides multispectral imagery across 13 spectral bands spanning Visible (VIS) to SWIR spectral ranges at spatial resolution of 10–60 m. Sentinel-2's multispectral instrument (MSI) includes two SWIR bands, Band 11 (~1560–1660 nm), and Band 12 (~2090–2290 nm), which coincide with two strong absorption features of the methane spectral profile. In particular, Band 12 contains the stronger absorption near 2300 nm, as compared to Band 11, making it more sensitive to the presence of methane. Although originally designed for land surface observations with lower spectral resolution (100–200 nm), Sentinel-2 has been shown to be capable of detecting large point source methane emissions by leveraging reflectance differences between these two spectral bands [10], [24], [32], [45]. In this study, we used all 13 bands from Sentinel-2 L1C top-of-atmosphere (TOA) reflectance data products, which are freely available to the user from the Copernicus Open Access Hub [48].

B. Dataset Construction

For this study, we compiled a series of datasets for the detection and segmentation of methane point sources using Sentinel-2 imagery. The pipeline illustrates the end-to-end process of generating these datasets, as shown in Fig. 1. The process begins with site selection based on previously confirmed methane-emitting infrastructure components (e.g., compressor stations, flares, pipelines) reported by Carbon Mapper (https://carbonmapper.org) and described in published literature (where possible). To ensure a broad generalization, sites were selected to represent a variety of landcover classes and ground surface characteristics. The Sentinel-2 L1C imagery acquired between January 2021 and June 2024 was then downloaded from the Copernicus Open Access Hub. For data reliability, we applied quality filtering to exclude cloudy (> 10% cloud cover) and low-quality acquisitions. Then each scene was cropped to 160 × 160 pixels for model input and converted from original GeoTIFF to 13-channel NumPy arrays corresponding to Sentinel-2 spectral bands at 20 m spatial resolution, thus preserving complete spectral information for model learning. The initial plume candidates were identified using the methane column enhancement map generated by the MBMP method [10]. These preliminary methane maps were then cross-validated with location-specific facilities and wind direction data from the ECMWF-ERA5 hourly reanalysis product [49] to confirm the presence of plumes. Finally, the verified plume locations were manually annotated to generate binary masks which represent the plume separated from the background pixels, thereby generating ground-truth labels for model training and evaluation.

The locations of our compiled image datasets for each target area are illustrated in Fig. 2. The dataset includes 44 methane point sources identified in differing land cover and operational settings. As shown in Fig. 2(a), these sources are concentrated in three major regions, the USA, Algeria, and Turkmenistan, with site types including compressor stations, flares, pipelines, and well-pad devices. Of these, 25 sites were confirmed as methane-emitting sources (positive), 14 were considered negative (no plumes were observed), and 5 were labeled undefined due to uncertain plume attribution. The spatial distribution of three main source groups can be seen in Fig. 2(b). Notably, the USA sites exhibit more complex surface features, including heterogeneous urban, agricultural, and vegetated areas, whereas the majority of sites in Algeria and Turkmenistan are located in unpopulated, semiarid regions with relatively homogeneous land cover. In addition, the temporal distribution of the observations varies across these

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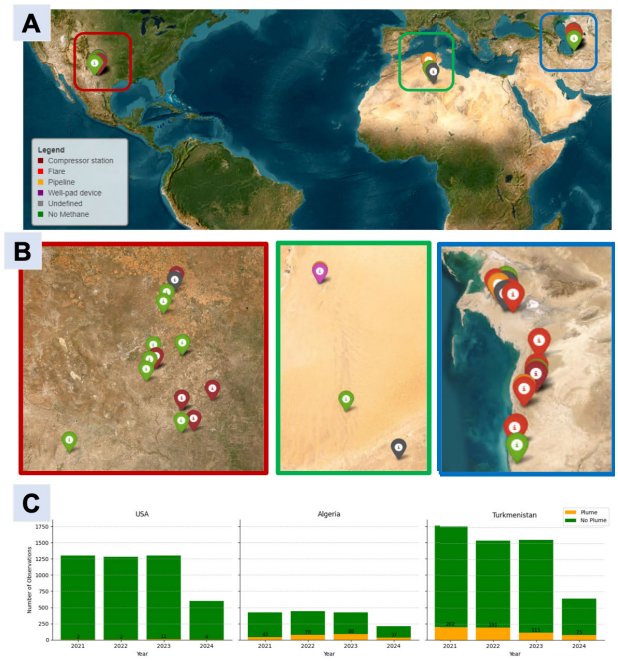


Fig. 2. Overview of spatial and temporal distribution of the methane point source dataset used in this study. (a) Locations of three target areas in USA (red box), Algeria (green) and Turkmenistan (blue) different types of land cover. (b) Locations of the 44 selected emission sites distributed across the three target areas shown and outlined in red, green and blue (a). (c) Plots showing the temporal coverage of Sentinel-2 observations in each of the target areas, acquired between 2021 and June 2024.

TABLE I

DATASET STATISTICS FOR TEMPORAL SPLIT SETTING (2021–2023 FOR TRAINING/VALIDATION AND 2024 FOR TESTING)

Dataset	Positive	Negative	Total
Training	594	7434	8028
Validation	134	1873	2007
Testing	147	1312	1459
Total	875	10619	11494

regions, as illustrated in Fig. 2(c). The USA contributes mainly negative samples, Algeria provides well-quantified and scaled plumes, while Turkmenistan supplies the most data and the majority of positive plume detections.

In total, the dataset comprises 11,494 samples, of which 875 (~8%) were labeled as methane plumes and the remaining 92% as nonplume samples. Each sample consists of: 1) a Sentinel-2 target date (TD) image (T) with all 13 spectral bands; 2) a Sentinel-2 reference date image ($T - 1$) with the same bands; 3) a binary plume mask. The dataset is divided as shown in Table I: all samples from 2024 are reserved for testing, while the remaining data are randomly split into training and validation datasets with an 8:2 ratio. This construction ensures diversity in both spatial and temporal domains, providing a robust foundation for model training and evaluation under different plume conditions.

Further to the above temporal splitting strategy used to evaluate temporal generalization, we construct a site-independent dataset split to assess the model's spatial generalization to unseen geographic locations. In this context, the dataset is partitioned based on geographic sites rather than acquisition time, as summarized in Table II. Specifically, the 44 observation sites are randomly partitioned into training (including validation) and testing sets at an approximate ratio of 5:1. All temporal observations associated with a given site are assigned exclusively to either the training or the testing set. This site-independent splitting strategy ensures that the evaluation rigorously reflects the model's spatial generalization capability. The corresponding results are reported in Section IV-D.

TABLE II

DATASET STATISTICS FOR SITE-INDEPENDENT SPLIT SETTING (NONOVERLAPPING GEOGRAPHIC SITES)

Region	Sites (Train+Val/Test)	Samples (Train+Val/Test)	Plume Samples (Train+Val/Test)
USA	14 / 3	3594 / 892	21 / 2
Algeria	3 / 1	1091 / 413	258 / 2
Turkmenistan	20 / 3	4699 / 805	522 / 70
Total	37 / 7	9384 / 2110	801 / 74

To assess the robustness of the proposed model beyond the constructed dataset, we collected additional small supplementary datasets from Khuzestan Province, Iran, an oil and gas producing region in Algeria, and a representative region in the United States. These regions were strictly excluded from the training, validation, and test splits and were used solely for out-of-domain generalization experiments. This design ensures a clear separation between in-domain and generalization evaluation, while enabling a realistic test of model transferability to previously unseen geographic regions.

III. METHODOLOGY

A. N-BPMSNet Model

The central innovation of N-BPMSNet lies in leveraging both temporal and spectral differences in methane plume segmentation, which provide complementary information for identifying dynamic methane plumes and enhancing their distinction from background features. Unlike conventional CNN-based methods, N-BPMSNet integrates physically informed spectral guidance with bitemporal input, enabling the network to capture the dynamic changes of the plume over time. The plumes are both transient and spatially constrained; hence, their spectral signature appears and disappears from certain pixels in certain images in the time series dataset. First, we explicitly incorporate the NDMI as an auxiliary input, defined as

$$NDMI = \frac{b_{11} - b_{12}}{b_{11} + b_{12}} \tag{1}$$

where b_{11} and b_{12} are spectral reflectance values in Sentinel-2 Bands 11 and 12. The NDMI provides a normalized measure of methane absorption properties in an image. Second, instead of relying solely on single-date images, N-BPMSNet adopts a dual-date strategy, comparing a TD image with a reference date (RD) image of the same location to capture the transience of the methane plume signature. The reference image is selected as the closest available cloud-free observation prior to the TD, ensuring minimal temporal separation while maintaining data quality. The key contribution of the bi-temporal approach is to capture the diagnostic physical property of

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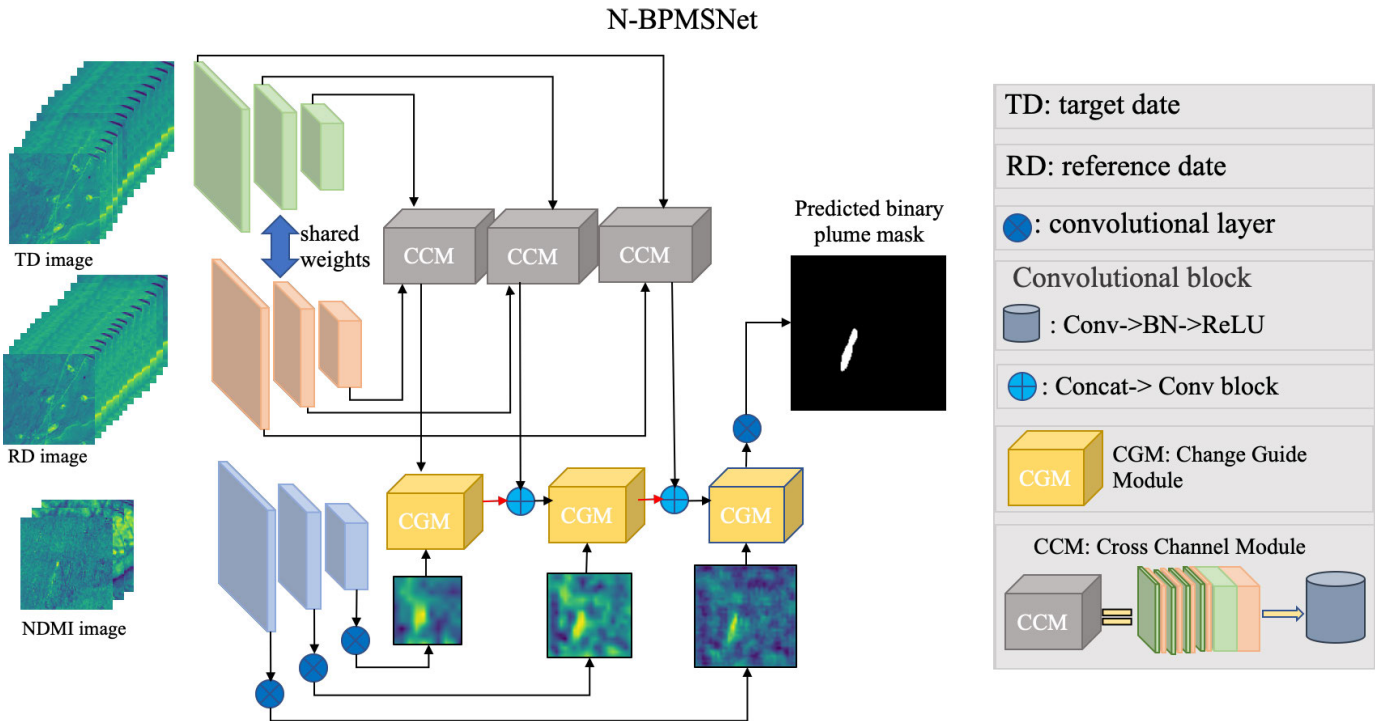


Fig. 3. Overview of the proposed N-BPMSNet structure. The model integrates dual-date inputs, an NDMI-guided branch, and specialized modules (CCM and CGM) within an encoder–decoder framework to generate accurate plume masks.

methane plumes as temporal varying features. In this case, a methane plume can be separated from those stationary features with similar spectral properties. Thus, "plume free" is not a requirement for a reference image while plume variation is essential. The bi-temporal framework captures these variations, emphasizing regions with significant spectral changes between the two observations. This enables the model to effectively identify methane emissions that may persist over time but fluctuate in concentration and position. This temporal comparison helps suppress background noise and atmospheric variations, enabling more effective discrimination of subtle plume signals.

The overall architecture follows an encoder–decoder structure, but unlike conventional networks that process only single input with uniform feature extraction, our design integrates moving, dual-date inputs (comparing image pairs) with shared weight encoders and introduces an NDMI-guided branch. As illustrated in Fig. 3, the TD and RD images are first encoded into spectral and spatial features, while the NDMI branch provides complementary spectral change information. Within the spectral encoder, a cross channel module (CCM) is embedded to capture interband correlations and encoder spectral absorption features into the latent feature space. The encoded features are then progressively refined by a change guide module (CGM), highlighting temporal and spectral variations based on NDMI information, fused in the decoder, and finally mapped into a binary plume mask. This design ensures that temporal, spatial and spectral patterns, NDMI guidance, and spectral dependencies are explicitly modeled, enabling more accurate and robust plume segmentation.

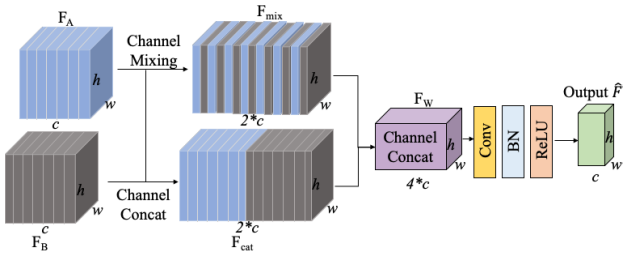


Fig. 4. Structure of the CCM, which performs crossconnection and concatenation of two feature maps for subsequent convolutional feature extraction.

B. Crosschannel Module

The CCM is designed to capture interband dependencies across multiple bands, as shown in Fig. 4. Given two feature maps from two dates F_A, $F_B \in \mathbb{R}^{C\times H\times W}$, the channel mixing is performed to obtain crosschannel interactions

$$F_{mix} = Mixing\,(F_A, F_B) \tag{2}$$

where F_A and F_B denote the feature maps at the target and reference dates, respectively, and C, H, and W are the number of channels, height and width of the feature maps. To preserve both the original spectral features and the mixed representation, these features are concatenated along the channel dimension

$$F_W = Concat\,(F_A, F_B, F_{mix}, F_{cat}) \tag{3}$$

where $F_{cat} \in \mathbb{R}^{4C\times H\times W}$. The fused feature is then processed by a convolutional block, including convolutional, batch normalization, and ReLU activation layers

$$\hat{F} = \sigma\,(BN\,(Conv\,(F_W))) \tag{4}$$

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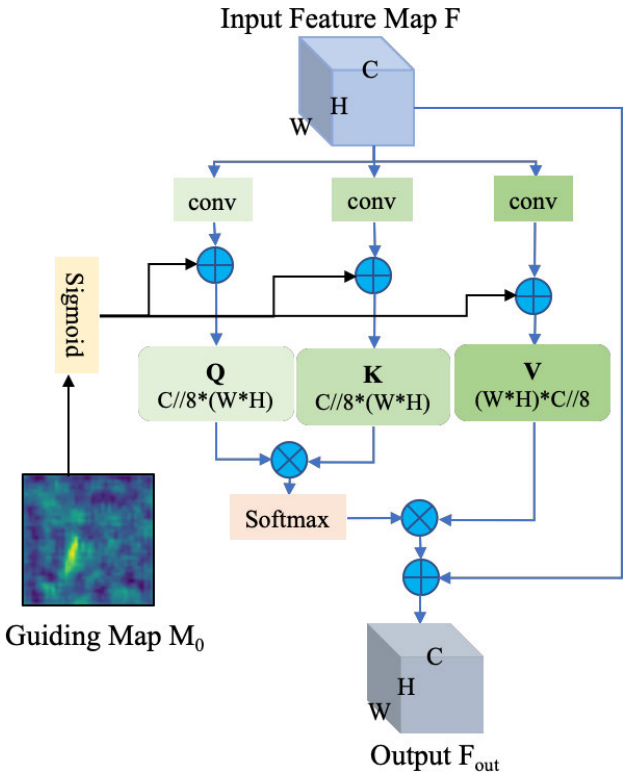


Fig. 5. Structure of the CGM. The guiding map is used to modulate the query, key, and value representations, followed by attention-based feature refinement and a residual connection to preserve global contextual information.

where σ denotes the ReLU activation function [50], $Conv(\cdot)$ represents a convolutional layer, and $BN(\cdot)$ denotes batch normalization. By integrating multitemporal and multispectral features, the module encodes spectral and temporal variation representations into the latent feature space. By stacking multiple CCMs at different scales, N-BPMSNet gradually strengthens spectral and temporal feature fusion and enhances plume segmentation accuracy.

C. Change Guide Module

The CGM, originally introduced in [51], is employed in our framework as a guidance branch to highlight temporally varying regions. CGM is designed to guide feature learning by emphasizing areas with significant temporal differences, thereby improving the localization of small structures. In N-BPMSNet, we adapt this module to operate on NDMI-based features, enabling it to specifically capture methane changes between the target-date and reference date images. While the core design follows [51], its integration into an NDMI-enhanced and dual-date pipeline is unique to our work, allowing CGM to serve as an additional guidance attention for plume segmentation.

As shown in Fig. 5, given an input feature map F and a guiding map M_0, the CGM first resizes and normalizes the guiding map

$$M = \sigma(M_0) \tag{5}$$

where σ denotes the sigmoid function. The guidance map is then used to modulate the feature representations by enhancing informative regions. Specifically, the query, key, and value are obtained through learnable projections and guided by M

$$Q, K, V = \left(W_Q * F,\ W_K * F,\ W_V * F\right) \odot (1 + M) \tag{6}$$

where W_Q, W_K, and W_V denote convolutional projection weights, and $\odot$ represents element-wise multiplication. The attention mechanism is then applied to capture long-range dependencies, and the output feature is computed with a residual connection

$$F_{\text{out}} = \gamma \cdot \text{Attention}(Q, K, V) + F \tag{7}$$

where γ is a learnable parameter implemented via a 1×1 convolutional layer and optimized through backpropagation during training. This design enables the model to emphasize regions with significant temporal variations while preserving global contextual information.

D. Loss Function

To handle severe class imbalance between plume and nonplume pixels, we designed a hybrid loss function which combines binary cross-entropy with logits (BCE) and Lovász-hinge loss. For a pixel i with ground truth label $y_i \in 0, 1$ and prediction z_i, the BCE loss is defined as

$$L_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \cdot \log\sigma(z_i) + (1 - y_i) \cdot \log(1 - \sigma(z_i))\right] \tag{8}$$

where $\sigma(z_i) = 1/(1 + e^{-z_i})$ denotes the sigmoid function and N is the number of pixels. The Lovász-hinge loss directly approximates the intersection-over-union (IoU) for binary segmentation. For ground truth labels $y \in -1, +1$ and predicted logits z, it is defined as

$$L_{LH} = \frac{1}{|y|}\sum_{j=1}^{|y|}\Delta_{IoU}(j) \cdot m_{(j)} \tag{9}$$

where $m_{(j)}$ are the margin errors sorted in pixel order $m_i = 1 - y_i z_i$, and $\Delta_{IoU}(j)$ denotes the discrete gradient of the IoU with respect to the sorted errors. It supervises more overlap between the predicted and "truth" plume area, especially for a small area of plumes. The overall loss function is then formulated as follows:

$$L = L_{BCE} + \lambda L_{LH} \tag{10}$$

where λ is a weighting factor that controls the contribution of the two classes. The weighting factor λ is introduced to balance the contributions of different loss components. Rather than being selected purely through performance tuning, λ is determined such that the magnitudes of the two loss terms are brought to a comparable scale during training. This design prevents one loss component from dominating the optimization process due to its larger numerical magnitude, thereby stabilizing training and ensuring balanced learning between detection and segmentation objectives. In practice, λ is set to 0.75 to approximately equalize their scales. We observed that slight variations around this value (e.g., 0.7 or 0.8) have minimal impact on overall performance, indicating that the model is not highly sensitive to the precise setting of λ.

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E. Evaluation Indicators

To comprehensively evaluate the performance of methane plume segmentation, both pixel-level metrics and scene-level metrics are employed in our research.

1) Pixel-Level Metrics: These metrics include Accuracy (Acc), balanced accuracy (BAcc), area under the receiver operating characteristic (ROC) curve (AUC), and IoU, which measure different aspects of model performance based on each pixel. Acc evaluates overall correctness, BAcc mitigates the influence of class imbalance, IoU quantifies the spatial overlap between predicted and ground-truth plume areas, and AUC reflects the classification ability independent of thresholds. Let TP, TN, FP, and FN be the numbers of true positives, true negatives, false positives, and false negatives for pixels. The metrics can be calculated by

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \tag{11}$$

$$BAcc = \frac{1}{2}\left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right) \tag{12}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{13}$$

$$AUC = \frac{1}{N_{\text{pos}} N_{\text{neg}}} \sum_{i=1}^{N_{\text{pos}}} \sum_{j=1}^{N_{\text{neg}}} \mathbf{1}\left(p_i^{\text{pos}} > p_j^{\text{neg}}\right) \tag{14}$$

where N_{pos} and N_{neg} denote the number of positive (plume) and negative (background) pixels, respectively; p_i^{pos} and p_j^{neg} denote the predicted probability of methane presence for the ith positive pixel and jth negative pixel; $\mathbf{1}(\cdot)$ is the indicator function, which equals 1 if the condition is true and 0 otherwise.

2) Scene-Level Metrics: Its evaluation assesses whether methane plumes are correctly identified in the entire images, reflecting the practicality of the model in real-world detection scenarios. Here, each "scene" refers to a 160×160 pixel image patch extracted from Sentinel-2 imagery. For each image, prediction results are binarized as plume or nonplume based on their overlap with the ground truth. In addition to Acc and BAcc, Precision (Prec), Recall (Rec) and $F1$-score ($F1$) are also applied to evaluate scene-level accuracy. The following metrics are reported, in which Acc and BAcc are the same as above, but defined at the scene level

$$Prec = \frac{TP}{TP + FP} \tag{15}$$

$$Rec = \frac{TP}{TP + FN} \tag{16}$$

$$F1 = \frac{2 \cdot Prec \cdot Rec}{Prec + Rec}. \tag{17}$$

By combining pixel-level and scene-level indicators, the evaluation framework provides a balanced assessment of both fine-grained segmentation accuracy and practical detection capability.

F. Implementation Details

All experiments were conducted on an NVIDIA A100 GPU with 40 GB of memory using the PyTorch framework. Input Sentinel-2 images were resampled at a spatial resolution of 20 m and cropped into nonoverlapping patches of size 160×160 pixels. Data augmentation techniques including horizontal/vertical flipping and rotations were applied during the training process. The network was trained using AdamW optimizer with an initial learning rate of $5e^{-4}$, weight decay of 0.0025, and a batch size of 16. A cosine annealing warm restart scheduler was employed, with the first restart period set to 50 epochs and subsequent periods increasing by a factor of 2. Training was performed for 500 epochs, and the model with the best IoU validation was selected for the final evaluation.

TABLE III
QUANTITATIVE COMPARISON WITH FOUNDATIONAL (PHYSICS-BASED) AND LEARNING-BASED METHANE PLUME DETECTION AND SEGMENTATION METHODS

Methods	Band Ratio	MBMP	CH4Net	ViT	N-BPMSNet
Pixel-level Metrics					
Acc	0.9788	0.9776	0.9988	0.9953	**0.9991**
BAcc	0.6314	0.6312	0.5608	0.6575	**0.7357**
AUC	-	-	0.9284	0.8687	**0.9936**
IoU	0.4946	0.4937	0.5519	0.5348	**0.6858**
Scene-level Metrics					
Acc	0.1069	0.0987	0.8938	0.7910	**0.9774**
BAcc	0.4581	0.4324	0.6631	0.6814	**0.9300**
Prec	0.0930	0.0882	0.4661	0.2516	**0.9014**
Rec	**0.8980**	0.8503	0.3741	0.5442	0.8707
F1	0.1685	0.1597	0.4151	0.3441	**0.8858**
AUC	-	-	0.8585	0.8397	**0.9856**
TP/FP	132/1288	125/1293	55/63	80/238	128/14
FN/TN	15/24	22/19	92/1249	67/1074	19/1298

IV. RESULTS AND DISCUSSIONS

A. Performance Against Existing Methods

To evaluate the effectiveness of the proposed N-BPMSNet, we first compare the performance of our method with current methods for methane detection and segmentation including Band Ratio [24], MBMP [10], CH4Net [46], ViT-based model (ViT) [45]. The first two are physics-based enhancement methods that do not generate binary masks. To enable a fair comparison, we applied a z-score threshold set at two standard deviations above the mean values, to convert their enhancement outputs into binary masks. Since these methods also lack confidence scores, the AUC cannot be reported. It should be noted that these physics-based methods also serve as the foundational approaches that enabled the spectral and temporal feature extraction in N-BPMSNet, and thus their results are included here as reference baselines rather than competing methods. In contrast, CH4Net and the ViT-based model are deep learning architectures that were reimplemented and trained on our dataset for fair comparison.

Quantitative comparison results in Table III demonstrate that N-BPMSNet consistently outperforms existing methods in both pixel-level and scene-level metrics. The physics-based Band Ratio and MBMP methods perform poorly on segmentation tasks because of the high sensitivity to surface reflectance and background variability, resulting in a large number of false positives and extremely low precision values (0.0930 and 0.0882, respectively), despite relatively high recall (0.8980 and 0.8503). This imbalance indicates that while these physical methods are capable of highlighting methane-related outliers,

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TABLE IV
PERFORMANCE COMPARISON OF N-BPMSNET WITH OTHER REPRESENTATIVE SEGMENTATION MODELS

Metrics \ Methods		Baseline	SiamCRNN	SwinSUNet	MambaBCD	N-BPMSNet
Pixel-level	Accuracy	0.9987	0.9988	0.9984	0.9988	**0.9991**
	Balanced Accuracy	0.6577	0.6541	0.6339	0.6991	**0.7357**
	AUC	0.9635	0.2963	0.0258	0.0985	**0.9936**
	IoU	0.6121	0.6133	0.5729	0.6433	**0.6858**
Scene-level	Accuracy	0.9657	0.8869	0.8547	0.8855	**0.9774**
	Balanced Accuracy	0.8299	0.8284	0.7380	0.8880	**0.9300**
	Precision	**1.0000**	0.4625	0.3640	0.4645	0.9014
	Recall	0.6599	0.7551	0.5918	0.8912	**0.8707**
	F1	0.7951	0.5736	0.4508	0.6107	**0.8858**
	AUC	0.9511	0.7469	0.1505	0.6572	**0.9856**
	TP/FP/FN/TN	97/0/50/1312	111/129/36/1183	87/152/60/1160	131/151/16/1161	128/14/19/1298

providing useful cues later exploited by our model, they lack discriminative ability to separate plumes from nonplume interference, leading to noisy detections and low $F1$-scores. In contrast, CH4Net and ViT achieve improved overall accuracy, but still suffer from false and missed detections. The ViT model has relatively higher recall (0.5442) but produces many false positives, while CH4Net ignores few true plumes, as reflected in its low recall (0.3741). By comparison, our proposed N-BPMSNet achieves the best performance across all evaluation metrics. The highest IoU (0.6858) represents its strong capability to accurately segment plume boundaries, even for small plumes. Both high recall and precision further confirm its robustness for weak and spatially subtle methane plumes that are often overlooked by existing approaches. The model also obtains superior balanced accuracy and $F1$-score, indicating robustness against class imbalance. Compared with existing methods, these results highlight the effectiveness of NDMI guidance and dual-date difference, enabling the model to achieve both high sensitivity and high reliability in methane plume segmentation while suppressing background noise.

B. Benchmarking General Segmentation Networks

We further benchmark N-BPMSNet against representative deep segmentation architectures, including UNet (Baseline) [38], SiamCRNN [39], SwinSUNet [40], and MambaBCD [41]. In the baseline setting, the UNet model takes both TD and RD images as input. All comparison models were trained and evaluated on our dataset in the same experimental settings. As illustrated in Table IV, the proposed N-BPMSNet achieves the best performance in all metrics with an accuracy of 0.9774, an AUC of 0.9856 and an $F1$-score of 0.8858. The UNet baseline has strong precision (1.0000) but relatively lower recall (0.6599), highlighting its shortcomings in missing some plumes. In particular, the baseline partially outperforms SiamCRNN, SwinSUNet, and MambaBCD, which can be attributed to two factors.

1) Transformer and sequential models such as SwinSUNet and MambaBCD typically require larger datasets to realize their advantages, whereas our dataset size is limited.
2) Convolutional architectures like UNet exploit local correlation and translation invariance, making them particularly effective for capturing localized spatial and spectral features in methane plume segmentation. In contrast, the design of N-BPMSNet by combining temporal and spectral features leads to more accurate plume localization and reduced false positives.

C. Performance Across Different Surface Types

To evaluate the robustness of the proposed model under varying surface complexities, we conduct a region-based performance analysis across the USA, Algeria, and Turkmenistan datasets. These regions represent different levels of surface heterogeneity, where the USA sites exhibit more complex and diverse backgrounds, while Algeria and Turkmenistan are dominated by relatively homogeneous semiarid environments. As shown in Table V, the proposed N-BPMSNet maintains consistently strong performance across all regions. At the pixel level, the model achieves comparable AUC values above 0.98 in all regions, indicating stable discrimination capability regardless of surface complexity. Although the IoU and balanced accuracy in the USA are slightly lower than those in arid regions, the performance remains competitive given the increased background complexity. At the scene level, the model achieves high accuracy and AUC across all regions. Notably, the recall in the USA reaches 0.8750, demonstrating the model's ability to effectively detect methane plumes even under challenging conditions. While the precision in the USA (0.7000) is lower compared to that in Algeria and Turkmenistan, this can be attributed to the more complex background and limited plume samples, which increase the likelihood of false positives. Overall, these results indicate that the proposed model is robust to surface heterogeneity and maintains reliable detection performance across regions with varying levels of complexity.

D. Spatial Generalization Evaluation

To further evaluate the spatial generalization capability of the proposed model, we conduct a site-independent experiment where the training and testing sets are constructed from distinct geographic locations using the site-independent split setting

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TABLE V
MODEL PERFORMANCE OF N-BPMSNET ACROSS REGIONS WITH VARYING SURFACE COMPLEXITY

Metrics\Regions	Pixel-level Metrics				Scene-level Metrics						
	Acc	BAcc	AUC	IoU	Acc	BAcc	Prec	Rec	F1	AUC	TP/FP/FN/TN
USA	0.9997	0.6617	0.9956	0.6365	0.9933	0.9350	0.7000	0.8750	0.7778	0.9941	7/3/1/590
Algeria	0.9981	0.7582	0.9897	0.7012	0.9533	0.9222	0.9362	0.8627	0.8980	0.9497	44/3/7/160
Turkmenistan	0.9987	0.7364	0.9922	0.6851	0.9705	0.9303	0.9059	0.8750	0.8902	0.9878	77/8/11/548

TABLE VI
SITE-INDEPENDENT EVALUATION WITH GEOGRAPHICALLY SEPARATE TRAINING AND TESTING LOCATIONS ACROSS DIFFERENT REGIONS

Metrics\Regions	Pixel-level Metrics				Scene-level Metrics						
	Acc	BAcc	AUC	IoU	Acc	BAcc	Prec	Rec	F1	AUC	TP/FP/FN/TN
USA	1.0000	0.9035	1.0000	0.8422	0.9978	0.9989	0.5000	1.000	0.6667	1.0000	2/2/0/888
Algeria	1.0000	0.9582	1.0000	0.7095	0.9927	0.9964	0.4000	1.0000	0.5714	0.9976	2/3/0/408
Turkmenistan	0.9987	0.6442	0.9315	0.6293	0.9714	0.8745	0.8983	0.7571	0.8217	0.9824	53/6/17/729
Total	0.9995	0.6596	0.9334	0.6419	0.9867	0.8824	0.8382	0.7703	0.8028	0.9828	57/11/17/2025

TABLE VII
ABLATION RESULTS OF DIFFERENT COMPONENTS IN N-BPMSNET

Metrics\Methods	Variants			Pixel-level Metrics				Scene-level Metrics					
	NDMI	CCM	CGM	Acc	BAcc	AUC	IoU	Acc	BAcc	Prec	Rec	F1	AUC
Baseline	×	×	×	0.9987	0.6577	0.9635	0.6121	0.9657	0.8299	1.0000	0.6599	0.7951	0.9511
M0	×	×	✓	0.9989	0.6610	0.9343	0.6252	0.9349	0.7826	0.7131	0.5918	0.6468	0.9297
M1	✓	×	×	0.9990	0.7045	0.9713	0.6607	0.9534	0.8895	0.7484	0.8095	0.7778	0.9535
M2	✓	✓	×	0.9990	0.7240	0.9899	0.6843	0.9472	0.8770	0.7160	0.7891	0.7508	0.9577
M3	✓	×	✓	0.9991	0.7228	0.9864	0.6821	0.9609	0.8726	0.8358	0.7619	0.7972	0.9678
N-BPMSNet	✓	✓	✓	**0.9991**	**0.7357**	**0.9936**	**0.6858**	**0.9774**	**0.9300**	**0.9014**	**0.8707**	**0.8858**	**0.9856**

dataset. The results are presented in Table VI. The proposed N-BPMSNet achieves consistent performance across different regions, including the USA, Algeria, and Turkmenistan. At the pixel level, the model maintains high accuracy and AUC across all regions. At the scene level, despite significant variation in plume sample numbers across sites, the model still achieves stable detection performance. Compared to the original temporal split, the overall performance shows slight variations (e.g., total pixel-level IoU decreases from 0.6858 to 0.6419), which is expected due to the unseen test sites. Importantly, the USA dataset contains more complex and heterogeneous surface conditions, yet the model maintains strong performance, demonstrating its robustness in challenging environments.

E. Ablation Study of Model Components

To analyze the contribution of individual modules, we perform ablation experiments by removing the CGM, CCM, or NDMI branch from the architecture. The baseline model employs only TD and RD images as input without NDMI, CCM, or CGM. Based on this baseline, four incremental variants were evaluated: M0, which only introduces the CGM module where the guided map is generated from feature maps of the first encoder layer; M1, which introduces the NDMI structure; M2, which further incorporates the CCM module; and M3, which integrates the CGM module instead of the CCM module, and N-BPMSNet is our proposed model. The results (Table VII) indicate that improvements are not uniform across all metrics, reflecting the complementary roles of the individual components. M0, which introduces only the CGM module, does not outperform the baseline at the scene level, indicating that CGM requires a well guided map to be effective. Compared with baseline, the M1 model achieves improved performance by importing the NDMI branch, which greatly reduces the missed detections and improves IoU and AUC from 0.6121 to 0.6607 and 0.9511 to 0.9535, respectively. The comparison between M1 and M2 shows that at the pixel level, CCM improves BAcc, IoU, and AUC, indicating enhanced spectral interactions. At the scene level, only AUC increases while F1, precision, or recall may slightly decrease, likely because CCM emphasizes global spectral consistency, which can slightly reduce detection of small or weak plumes. This highlights CCM's complementary role in N-BPMSNet, mainly enhancing pixel-level segmentation. Furthermore, the difference between M1 and M3 highlights the advantage of the CGM module with the effective NDMI-guided map to more accurately localize temporally varying plume regions, resulting in considerable improvements in IoU from 0.6607 to 0.6821, $F1$-score from 0.7778 to 0.7972, and AUC from 0.9535 to 0.9678. Overall, the ablation results confirm that each component contributes differently to the overall performance, with NDMI providing the most critical guided features, CCM enhancing spectral interaction and feature consistency, and CGM complementarily improve spectral and temporal

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TABLE VIII
PERFORMANCE COMPARISON BETWEEN SINGLE-TEMPORAL (TD ONLY) AND BI-TEMPORAL (TD + RD) INPUTS. (TD: TARGET DATE IMAGE, RD: REFERENCE DATE IMAGE)

Metrics\Models	Input		Pixel-level Metrics				Scene-level Metrics						
	TD	RD	Acc	BAcc	AUC	IoU	Acc	BAcc	Prec	Rec	F1	AUC	TP/FP/FN/TN
Baseline(Single)	✓	×	0.9988	0.5608	0.9284	0.5519	0.8938	0.6631	0.4661	0.3741	0.4151	0.8585	55/63/92/1249
N-BPMSNet(Single)	✓	×	0.9989	0.6130	0.8924	0.5968	0.9308	0.7350	0.7347	0.4898	0.5878	0.8808	72/26/75/1286
Baseline	✓	✓	0.9987	0.6577	0.9635	0.6121	0.9657	0.8299	1.0000	0.6599	0.7951	0.9511	97/0/50/1312
N-BPMSNet	✓	✓	0.9991	0.7357	0.9936	0.6858	0.9774	0.9300	0.9014	0.8707	0.8858	0.9856	128/14/19/1298

localization and accuracy, respectively. The full integration of all components in N-BPMSNet achieves the best performance across metrics.

While the ablation study evaluates the contribution of individual modules under the bi-temporal setting, we further examine the importance of temporal information by comparing models with single-temporal and bi-temporal inputs. Table VIII presents the results of models using either the target image (TD) alone or jointly with the reference image as input, where "Baseline (Single)" and "N-BPMSNet (Single)" denote models operating on TD only. The results indicate that both models with single-temporal input exhibit significantly poorer performance than those with bi-temporal inputs, suggesting the importance of temporal information in capturing dynamic spectral variations associated with methane plumes. Moreover, even under the single-temporal setting, N-BPMSNet (Single) outperforms the baseline, indicating improved feature representation capability. When extended to the bi-temporal setting with NDMI guidance, the full N-BPMSNet achieves the best performance, highlighting the effectiveness of jointly exploiting temporal, spectral, and spatial information.

F. Qualitative Evaluation and Visualization

Representative visualization results are presented in Fig. 6, which shows the results of different models in 15 randomly selected test cases that cover both heterogeneous and homogeneous surface types. Each row corresponds to a specific case, while the columns compare the results of traditional band ratio and MBMP methods, existing methane plume networks (CH4Net, ViT), general segmentation architectures (Baseline, SiamCRNN, SwinSUNet, MambaBCD), our ablation variants (from N-BPMSNet_v0 to N-BPMSNet_v3), and the proposed N-BPMSNet. Across different surface conditions, traditional Band Ratio and MBMP methods are often easily affected by interference from the spectral signatures of vegetation and water, and objects with low surface albedo resulting in false detections, as shown in the 3rd, 10th, and 14th rows. Although deep segmentation models capture plume areas more consistently and improve plume localization accuracy, they tend to miss plumes with smaller emissions and often misclassify background changes as plumes due to noise and atmospheric changes. In contrast, N-BPMSNet achieves the most accurate segmentation, producing spatially consistent plume masks that closely align with ground truth. Our model maintains plume continuity against homogeneous backgrounds and avoids false detections in challenging heterogeneous scenes. These improvements are particularly noticeable in the case of small or elongated plumes, which other methods either fragment or suppress entirely.

To further interpret the prediction process of our network, we employed gradient-weighted Class Activation Mapping (Grad-CAM) [52] to generate heatmaps that visualize the spatial regions most strongly influencing the network's predictions (Fig. 7). These heatmaps were derived from the activation maps of the final convolutional layer, which capture the most semantically meaningful spatial features for methane plume segmentation. The visualizations present qualitative results in several representative cases of methane plumes across different types of land cover. The plume-relevant regions highlighted in the heatmap activations align well with the plume binary mask, while effectively suppressing background interference. These results suggest that the model focuses on physically meaningful spectral and spatial regions rather than on false background information, thereby enhancing the reliability of methane plume detection.

G. Model Validation: Case Studies on Regions in Iran, Algeria, and USA for Unknown Methane Emissions

To further evaluate the generalizability of N-BPMSNet, we evaluated its performance in three regions that were not included in the training or validation datasets. Khuzestan Province, Iran, an oil and gas production area in Algeria, and a representative region in the United States. These case studies serve as practical tests of the generalization ability of the model under diverse geographic and atmospheric conditions.

1) Case Study of Iran: As shown in Fig. 8, the model was applied to Sentinel-2 imagery from Khuzestan Province, Iran. N-BPMSNet successfully detected methane plumes on multiple dates in 2024, closely aligning with the Band Ratio visualizations. The detected plumes exhibited varying directions and shapes depending on weather conditions on the day, and were generally consistent with the prevailing wind direction, demonstrating the model's ability to accurately capture plume morphology under various atmospheric conditions. In some cases, slight deviations from the wind field can be observed, particularly under low wind speed conditions, where methane may accumulate and diffuse around the emission source. Additionally, this discrepancy can be attributed to the use of ERA5 reanalysis data, which represents spatially and temporally averaged wind fields at 10 m height rather than local instantaneous conditions at the plume transport altitude. Despite these effects, the temporal sequence of the detections

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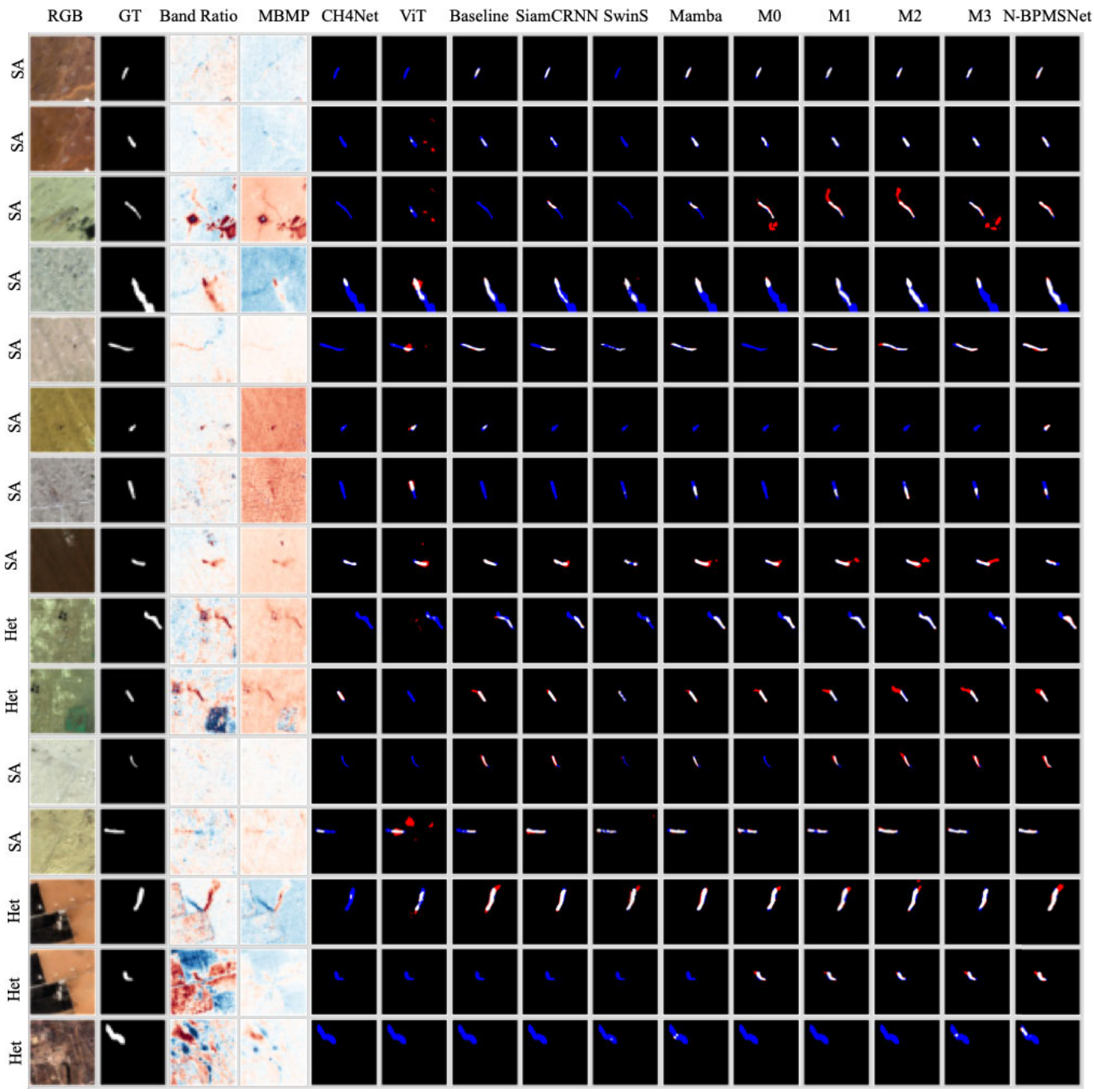


Fig. 6. Visualization of methane plume segmentation results on different cases. TP (white), TN (black), FP (red), and FN (blue). GT denotes ground truth, SwinS denotes SwinSUNet, mamba denotes MambaBCD, and M0–M3 denote different variants of N-BPMSNet. The labels on the left indicate the dominant land type of each case: SA = semiarid, het = heterogeneous environment.

in these complex surface areas further suggests the presence of a persistent leak source, highlighting the value of N-BPMSNet in monitoring long-term emissions.

2) Case Study of Algeria: N-BPMSNet was applied to an oil and gas production area (31°40′12.0″N 6°11′24.0″E) in Algeria. As illustrated in Fig. 9, the model successfully identified methane plumes at multiple sites (labeled A-G) and different time periods. The predicted masks (left panels) consistently matched the plume patterns observed in the Band Ratio visualizations (right panels), validating the reliability of the detections. The results showed that our model could accurately distinguish true methane plumes from irrelevant signals, including smoke from burning flares, cloud shadows, and dark soil backgrounds. Furthermore, the detections revealed two distinct types of emission source. First, the model consistently captured persistent emission hotspots at sites A, C, and E, where plumes were observed repeatedly over multiple dates. These patterns suggest ongoing leakage from industrial facilities. Second, N-BPMSNet identified transient point source events at sites B, D, F, and G, where plumes

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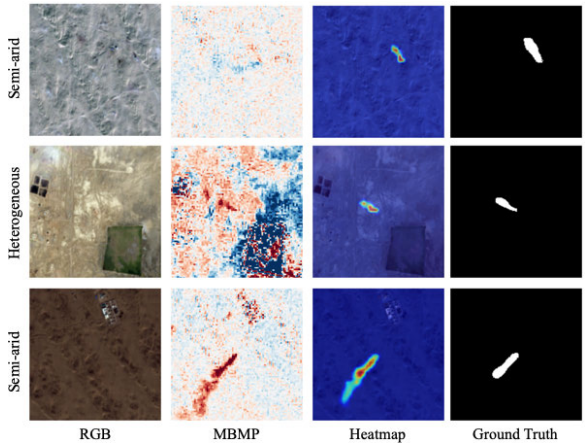


Fig. 7. Heatmap visualization of methane plume detection using N-BPMSNet. (Left column) RGB images derived from Sentinel-2 observation. *Unless otherwise stated, all RGB images in this study are true-color composites generated from Sentinel-2 bands 4 (red), 3 (green), and 2 (blue).*

only occurred once during the observations. The ability to detect both persistent and transient emissions underscores the sensitivity of the model to diverse plume behaviors.

3) Case Study of USA: To further evaluate performance under complex surface conditions, we conduct a case study in a heterogeneous region in the United States, including industrial, agricultural, and natural land covers. As shown in Fig. 10, three representative subregions (A–C) are detected with plumes under different wind conditions. N-BPMSNet successfully captures methane plumes of varying intensity and morphology, including weak and elongated structures, while effectively suppressing background interference. Predicted plumes closely match the Band Ratio highlights, confirming reliable localization even under complex backgrounds.

In general, these case studies demonstrate that N-BPMSNet is generalized effectively to unseen regions with complex background characteristics. The model not only adapts to variations in plume shapes driven by wind fields but also remains robust against surface artifacts. Importantly, it captures both persistent emitters and transient emission events, offering comprehensive insights into regional methane emission dynamics. These results highlight the robustness and generalization of N-BPMSNet, providing strong evidence for its potential for long-term monitoring and identification of methane sources on a global scale.

H. Computational Complexity Analysis

To further evaluate the practical applicability of the proposed method, we analyzed the computational complexity and efficiency of N-BPMSNet in terms of parameter size, FLOPs, and inference performance. Table IX presents a comparison of model complexity and inference efficiency across different methods. The proposed N-BPMSNet contains only 1.24 million parameters, which is significantly fewer than other deep learning models such as SiamCRNN (28.54M), SwinSUNet (42.96M), and MambaBCD (85.53M). In terms of computational cost, N-BPMSNet requires only 1.29 GFLOPs, indicating a lightweight architecture. Despite its compact design, the inference time remains highly efficient (1.7 ms per sample), comparable to the baseline and CH4Net. The results demonstrate that N-BPMSNet achieves a favorable balance between computational efficiency and detection performance.

TABLE IX
COMPARISON OF MODEL COMPLEXITY AND INFERENCE EFFICIENCY ACROSS DIFFERENT METHODS

Model	Params (M)	FLOPs (G)	Inference Time (ms)
Baseline	13.59	17.23	1.7
CH4Net	13.59	17.04	1.7
ViT	20.94	17.31	1.8
SiamCRNN	28.54	11.15	5.2
SwinSUNet	42.96	12.12	17.1
MambaBCD	85.53	4.57	35.8
N-BPMSNet	1.24	1.29	1.7

In terms of large-scale efficiency, N-BPMSNet processes thousands of $\sim 3 \times 3$ km Sentinel-2 image tiles within a few minutes on a single NVIDIA A100 40GB GPU. Extrapolating to the continental United States (approximately 1.09 million tiles), the estimated processing time on a single GPU is about 24 h. Utilizing multiple GPUs can proportionally reduce runtime; with four GPUs, the estimated processing time is approximately six hours, demonstrating that N-BPMSNet is practical for near real-time, large-scale methane plume monitoring.

I. Limitations

Although N-BPMSNet achieves robust performance in most scenes, several representative failure cases are illustrated in Fig. 11. Specifically, two typical situations can be identified: 1) missed detections of weak plumes with low concentration, where probability responses remain below the detection threshold (first row) and 2) low-quality imagery with spectral artifacts reducing reliability (second row). These errors are largely attributable to factors such as plume emission concentration levels and data quality, reflecting external constraints on performance.

In addition to these failure cases, two special scenarios are worth discussing: offshore and densely vegetated areas, where methane emissions exist but plume detection is particularly challenging. In both types of environment, the spectral and illumination complexities are different from typical barren land surface conditions, thereby affecting the detection performance, as illustrated in Fig. 12 with two examples in these special scenarios. For offshore regions, plume detection is complicated by the specular reflection of the water's surface. However, our results indicate that N-BPMSNet can still successfully capture methane plumes over open water. In contrast, the vegetation case (bottom row), with plume annotations provided by Carbon Mapper [53], highlights a limitation: the model fails to detect plumes reliably over densely vegetated areas. The spectral signature of vegetation in the two SWIR bands of Sentinel-2 is similar to that of methane at these wavelengths. Due to the micro-scale bandwidths spectral resolution of Sentinel-2 (and most multispectral sensors), the model fails to consistently detect methane emissions

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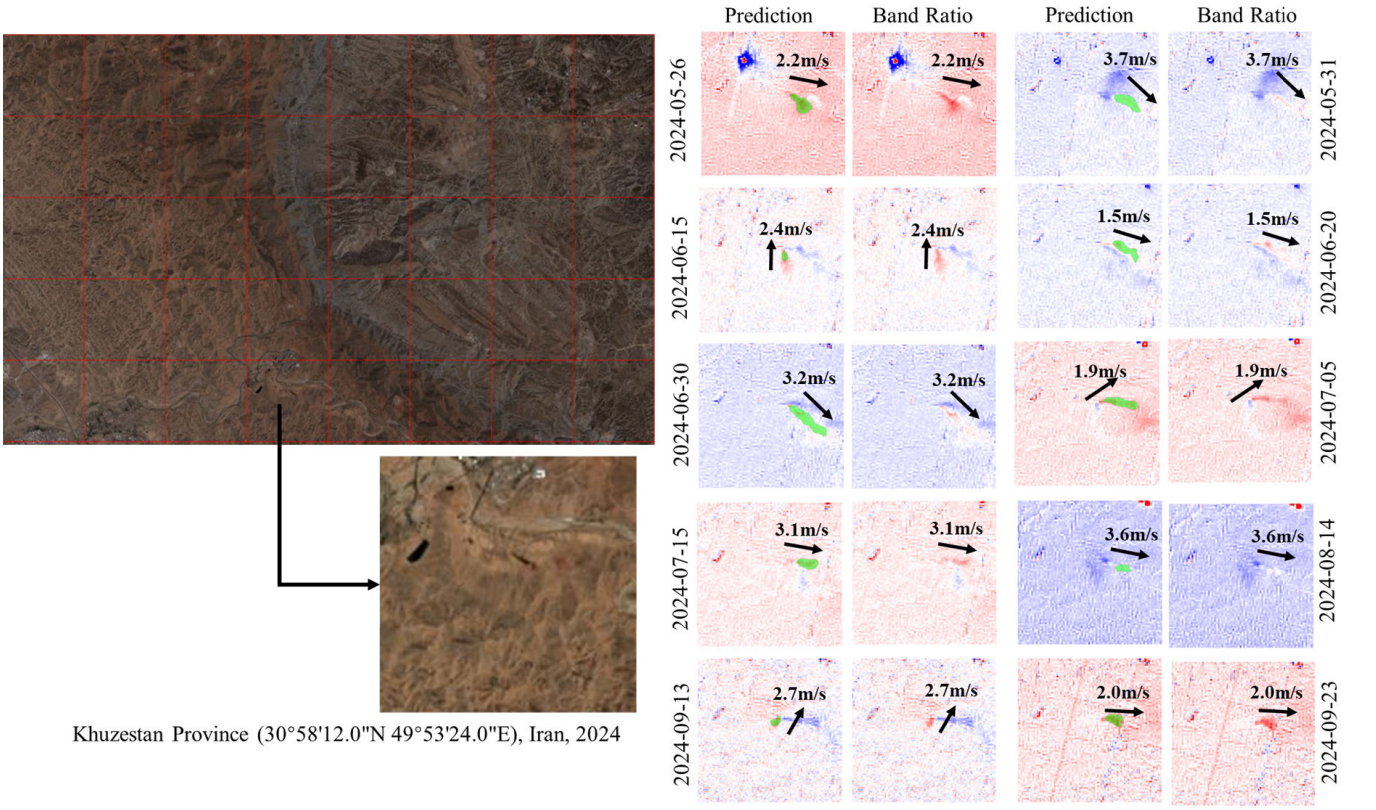


Fig. 8. Out-of-domain generalization results of N-BPMSNet applied to khuzestan Province, Iran (2024) (unseen region; not part of training/validation/test datasets). (Left) Sentinel-2 RGB images of the study area with detected methane plume regions highlighted. (Right) corresponding predicted plume masks by N-BPMSNet and band-ratio visualizations derived from sentinel-2 data. In the visualizations, red indicates areas of high methane enhancement, green marks the model's predictions, and blue represents lower background methane enhancement. Dates and estimated wind speeds are annotated. The model consistently detects methane plumes across multiple observation dates, highlighting persistent emission sources in the region.

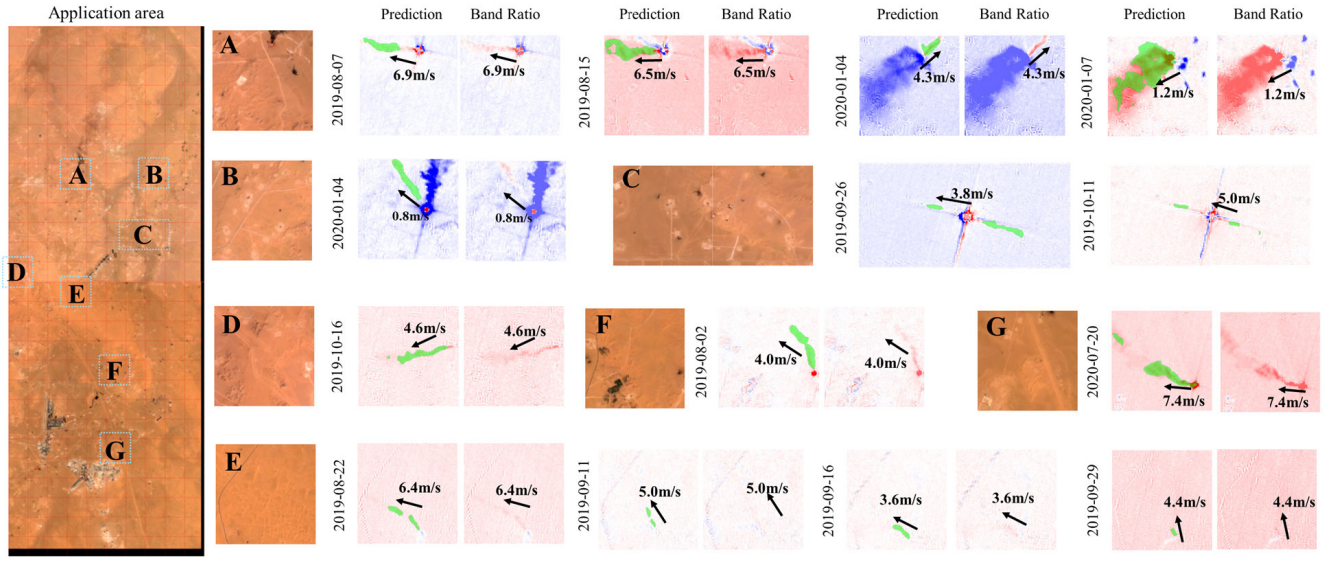


Fig. 9. Out-of-domain generalization results of N-BPMSNet applied to an oil- and gas-producing region in Algeria (unseen region; not part of training/validation/test datasets). (Left) Sentinel-2 RGB images of the application area with sites labeled A–G. (Right) predicted plume masks and corresponding band-ratio visualizations across multiple dates, with estimated wind speeds annotated. In the visualizations, red indicates areas of high methane enhancement, green marks the model's predictions, and blue represents lower background methane enhancement. The model accurately detects plumes of varying scales and orientations, highlighting recurrent emission events in the region.

above dense vegetation. This limitation is not due to the network design itself, but originates from the intrinsic spectral similarities between vegetation and methane, and the limited spectral resolution of Sentinel-2. Future work will address

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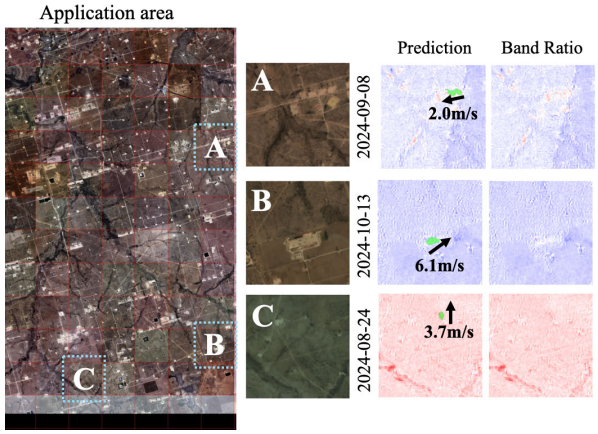


Fig. 10. Out-of-domain generalization results of N-BPMSNet applied to a complex onshore region in the United States (unseen region; not part of the training/validation/test datasets). (Left) Sentinel-2 RGB image of the study area, with representative subregions labeled A–C. (Right) Corresponding band-ratio visualizations and predicted plume masks under different wind conditions. The arrows indicate wind direction, with wind speeds annotated for each case. In the visualizations, red indicates areas of high methane enhancement, green marks the model's predictions, and blue represents lower background methane enhancement.

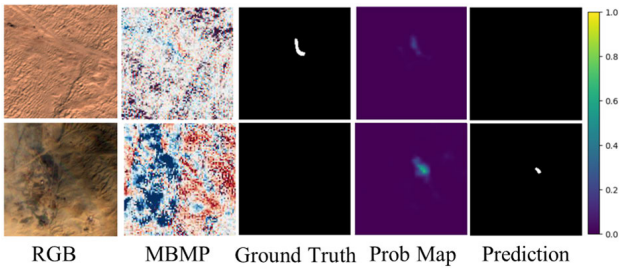


Fig. 11. Representative failure cases of N-BPMSNet. (Left) Each row shows an example true color composite RGB image, MBMP visualization, ground-truth annotation, prediction probability map, and final segmentation (prediction). The prediction probability map represents the pixel-wise likelihood of methane plume presence, obtained directly from the network output after sigmoid activation.

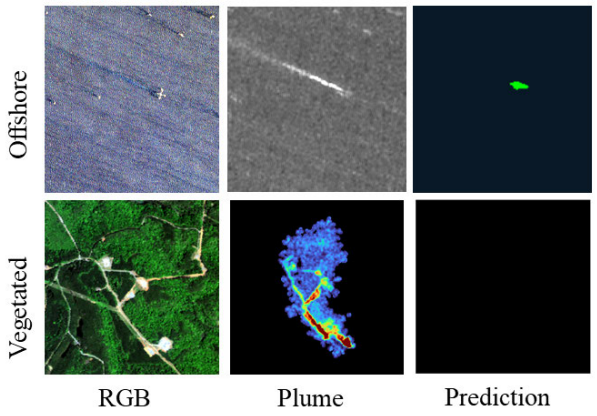


Fig. 12. (Top row) Special case analysis of offshore and (bottom row) vegetated areas for methane plume detection. (Columns from left to right) RGB images based on Sentinel-2, corresponding plume maps (ground truth), and predicted plume masks by N-BPMSNet. The offshore plume was generated using ERMapper, while plume in the vegetated area was obtained from carbon mapper.

this limitation by integrating hyperspectral imagery to enhance methane detectability in densely vegetated areas.

V. Conclusion

In this study, we propose N-BPMSNet, a Sentinel-2–based framework for automated methane plume detection and segmentation that integrates differential temporal, spectral and spatial information. Extensive experiments on a large-scale real-world dataset that we constructed and case studies in diverse regions have confirmed the accuracy, robustness, and strong out-of-domain generalization capacity of the proposed model. In particular, N-BPMSNet effectively handles heterogeneous surfaces and shows clear advantages in detecting small or elongated plumes, which are often fragmented or entirely suppressed by existing methods. Moreover, it captures both persistent emitters and transient emission events, providing comprehensive insights into regional methane emission dynamics. Beyond detection accuracy, the framework also demonstrates computational efficiency, processing over a thousand 3×3 km Sentinel-2 tiles in under two minutes on a single GPU, and scaling efficiently to nation-wide monitoring scenarios. These findings confirm that N-BPMSNet has strong practical utility for scalable and detailed methane monitoring, supporting more effective and targeted emission mitigation. Importantly, by leveraging freely available Sentinel-2 imagery, N-BPMSNet fills a critical open-access gap in methane monitoring, offering a cost-effective alternative to commercial sensors and services that are otherwise restrictive or expensive. Future work will explore multisensor integration and advanced spectral modeling to further enhance detection under complex environmental conditions.

Acknowledgment

More information about the Resource Geophysics Academy can be found at: https://www.imperial.ac.uk/resource-geophysics-academy/

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